# Kartik Umesh Chavan

**Email:** kartikchavan208@gmail.com
**Mobile:** +91 7021222457
**LinkedIn:** linkedin.com/in/kartikchavan01
**GitHub:** github.com/Chavan-Kartik

## Education

- **Vishwakarma Institute of Technology** — Pune, India
  *B.Tech in Computer Science & Engineering (AI & ML); CGPA: 8.74* — *Expected May 2027*

## Technical Skills

- **Languages**: C++, Python, JavaScript, TypeScript, SQL, HTML/CSS
- **Frameworks**: React.js, FastAPI, Flask, Tailwind CSS, Redux, Node.js, , Generative AI Workflows
- **Infrastructure & Tools**: Docker, Redis, PostgreSQL, MySQL, Postman, WebSockets, Winsock2, Gemini API
- **Concepts**: Distributed Systems, RESTful APIs, Concurrency, Object-Oriented, Programming (OOP)

## Experience

- **Industrial Training and Assessment Center (ITAC) - OSU** — Oklahoma, USA (Remote)
  *AI/ML Engineering Intern (The US Dept. of Energy - Funded Project)* — *Feb 2026 – March 2026*
  - Engineering a secure, air-gapped **Zero-Trust RAG pipeline** (**Llama 3, ChromaDB, LangChain**) to process proprietary DOE audits.
  - Building automated workflows using generative AI to instantly convert technical data into adaptive, multi-modal educational content.
- **Canspirit AI** — Remote/Hybrid
  *UI/UX & Generative AI Intern* — *July 2025 – Nov 2025*
  - Optimized asset creation workflows for a E-commerce website by integrating AI tools like **Gemini Pro** and **Nano Banana**, significantly reducing image production turnaround time.
  - Engineered **system prompts** ensuring stylistic consistency across UI assets, and leveraged AI alongside Photoshop/Illustrator to enhance design efficiency.

## Key Projects

- **CastOS: AI-Powered Casting Platform (Industry Project)**: Architected a neuro-symbolic platform solving the casting Knapsack Problem via **Reinforcement Learning (PPO)**, achieving 92% budget adherence. Implemented a **Zero Trust** security model and reduced semantic extraction latency through content-hashed caching strategies.
  **Tech Stack:** React, FastAPI, Redis, Docker, Clerk Auth, Gemini API
- **Low-Latency Lock-Free Pub-Sub Broker**: Engineered a high-throughput messaging broker handling persistent TCP clients with **deterministic sub-millisecond latency**. Eliminated critical-path locks using an **MPMC concurrent queue** and implemented a custom **fixed-size memory pool** to prevent heap fragmentation.
  **Tech Stack:** C++, Winsock2, Systems Programming, Concurrency

## Leadership & Achievements

- **Campus Lead, GDG on Campus, VIT Pune (2025-26)**: Leading a team of 125+ students. Achieved **Tier 1 status in India** for Google Cloud Study Jams, facilitating **100+ certifications**. Hosted **Asia's first Liquid Galaxy Lab** and organized **"Hack-O-Verse,"** managing logistics and sponsorship for thousands of developers.
- **Rajya Puraskar Awardee**: Received the prestigious state Rajya Puraskar Award in Scouting.
- **Performer (Band Leader)**: Lead vocalist; performed in front of audiences exceeding 1,000+ attendees.
- **Volunteering & Social Impact**: Served as a **Financial Educator** (iVolunteer) for women's financial literacy, and an **Utkarsh Volunteer** (VIT) mentoring underprivileged children.
- **DevX AI/ML Trainer**: Led a session on Recurrent Neural Networks (RNNs), explaining architecture and linking to Transformers.